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Section B

Answer **all** *the questions in the spaces provided.*

2. A class of students were using dice to model radioactive decay. There were 10 groups of students in the class. Each group of students had 90 dice.

(a) (i) Describe how they should carry out the experiment. [3]

....

....

....

....

....

(ii) State the number of dice you would expect to land with a 6 facing up on the first throw in **each** group. [1]

....

(iii) Each group's results were added together to give the class results. Give **one** advantage of using a bigger sample size. [1]

....

....

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(b) The table shows the results for the **whole class**.

Throw	Number of sixes	Number of dice remaining
0	0	900
1	146	754
2	124	630
3	103	527
4	....	442
5	72	370
6	61	309
7	52	257
8	38	219
9	34	....

(i) **Complete** the table. [2]

(ii) Use the data in the table to estimate the half-life of the 'dice'. [1]

half-life = ... throws

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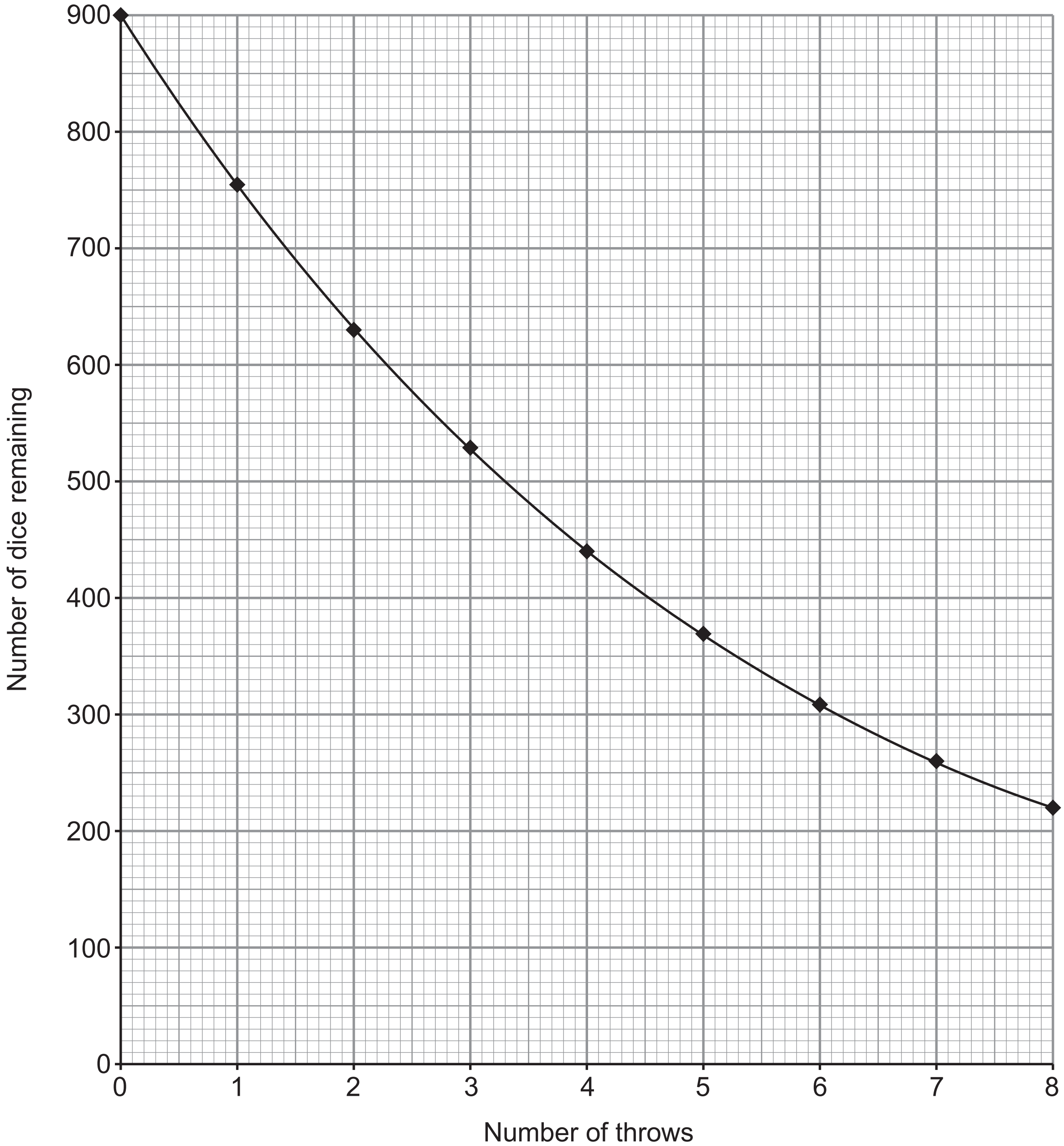
(*c*) Some of the data in the table is plotted on the grid below.



(i) **Draw lines** on your graph to find the half-life of the dice. [2]

half-life = ... throws

(ii) State **one** reason why finding the half-life using the graph is a better method than the one used in part (*b*)(ii). [1]

....

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(d) Radioisotopes have many medical uses. One use is to sterilise blood products for transfusion.

(i) Select from the table below the most suitable radioisotope for this purpose. [1]

....

Radioisotope	Half-life	Type of emissions
bismuth-213	46 minutes	alpha
chromium-51	28 days	gamma
caesium-137	30 years	beta and gamma
bismuth-209	2×10^{19} years	alpha

(ii) Give **two** reasons for your choice. [2]

1. ....

2. ....

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